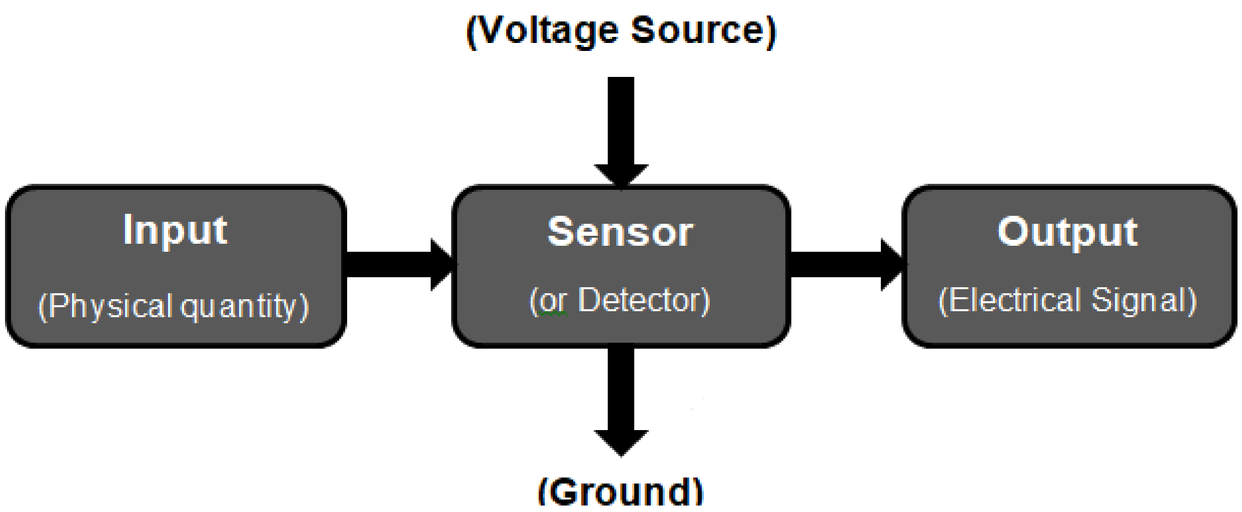
Unit - 3
Sensors and Actuators

Objective

- Sensors and its Applications
- Types of Sensors
- Discuss Different types of sensors and its use
- What is Actuators
- Applications of Actuators
- Discuss Different types of Actuators and its use

Introduction to Sensors

- The sensor can be defined as a device which can be used to sense/detect the physical quantity like force, pressure, strain, light etc and then convert it into desired output like the electrical signal to measure the applied physical quantity



Characteristics of Sensors

Less Noise and Disturbance

Less power consumption

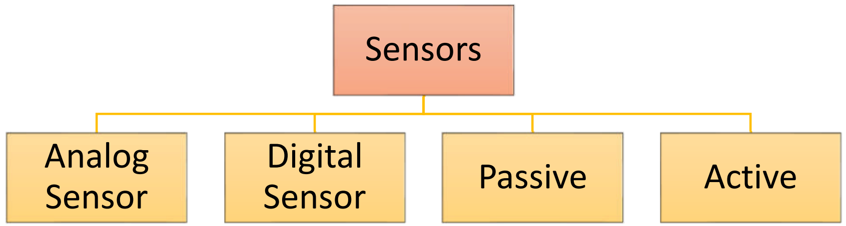
High Sensitivity

High Resolution

Linearity

Types of Sensors

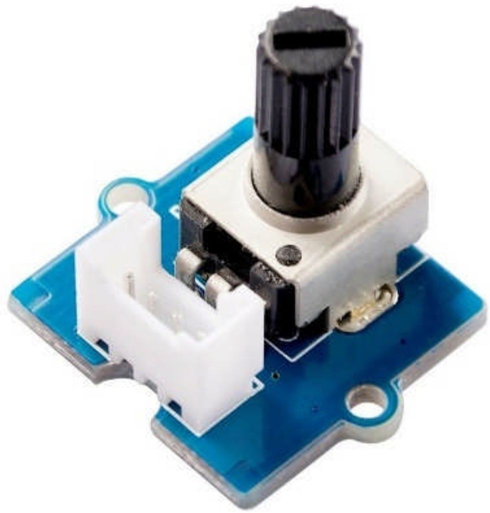
- **Analog Sensor** : The sensor that produces continuous signal with respect to time with analog output is called as **Analog sensors**
- **Digital Sensor** : When data is converted and transmitted digitally, it is called as **Digital sensors**. Digital sensors are the one, which produces discrete output signals



- **Active sensors** are those which do not require external power source for their functioning
- **Passive sensors** require external power source for their functioning.

Rotary Angle Sensor - Potentiometer

- The rotary angle sensor produces analog output between 0 and Vcc (5V DC with Seeeduino) on its D1 connector. The D2 connector is not used.
- The angular range is 300 degrees with a linear change in value.
- The resistance value is 10k ohms, perfect for Arduino use. This may also be known as a "potentiometer".



Sound Sensor

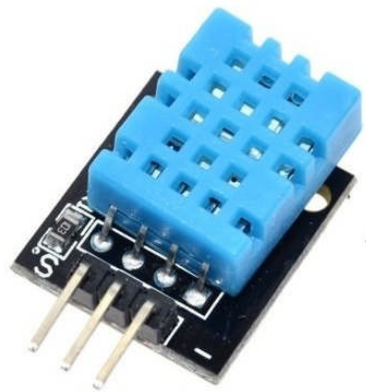
- The sound sensor is one type of module used to notice the sound.
- Generally, this module is used to detect the intensity of sound.
- The applications of this module mainly include switch, security, as well as monitoring.
- The accuracy of this sensor can be changed for the ease of usage.
- This sensor is capable to determine noise levels within DB's or decibels at 3 kHz 6 kHz frequencies approximately wherever the human ear is sensitive

- Pin1 (VCC): 3.3V DC to 5V DC
- Pin2 (GND): This is aground pin
- Pin3 (DO): This is an output pin

Application of Sound Sensor

- Security system for Office or Home
- Spy Circuit
- Home Automation
- Robotics
- Smart Phones
- Ambient sound recognition
- Audio amplifier
- Sound level recognition (not capable to obtain precise dB value)

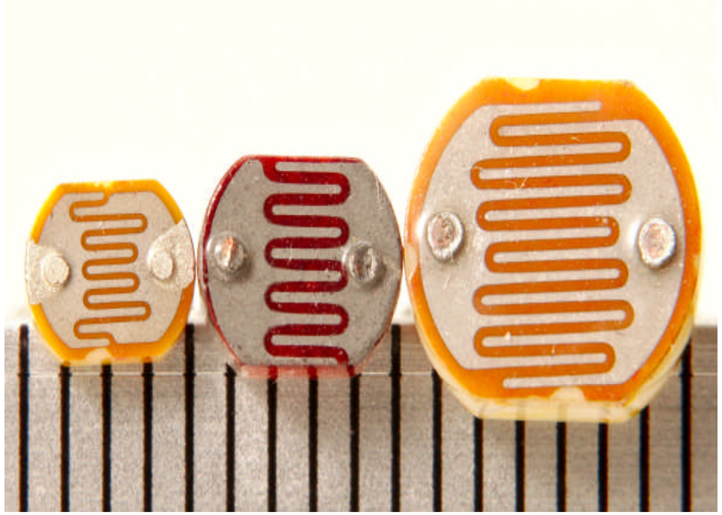
Temperature Humidity Sensor



- Temperature and humidity sensor (or rh temp sensor) is devices that can convert temperature and humidity into electrical signals that can easily measure temperature and humidity.
- DHT11 is a Humidity and Temperature Sensor, which generates calibrated digital output. DHT11 can be interface with any microcontroller like Arduino, Raspberry Pi, etc.
- Only three connections are required to be made to use the sensor - Vcc, Gnd and Output.

Light Sensor

- A light sensor is a photoelectric device that converts light energy (photons) detected to electrical energy (electrons)
- The light sensors used in robots are two types photovoltaic cells & photoresistors.
- Photovoltaic cells are used to change the solar radiation energy to electrical and these sensors are used in solar robot manufacturing.
- Photoresistors are used to alter their resistance by modifying light intensities.
- LDR is one the sensor belongs to light sensor family.



LDR – Light dependent Resistors

Ultrasonic Sensor

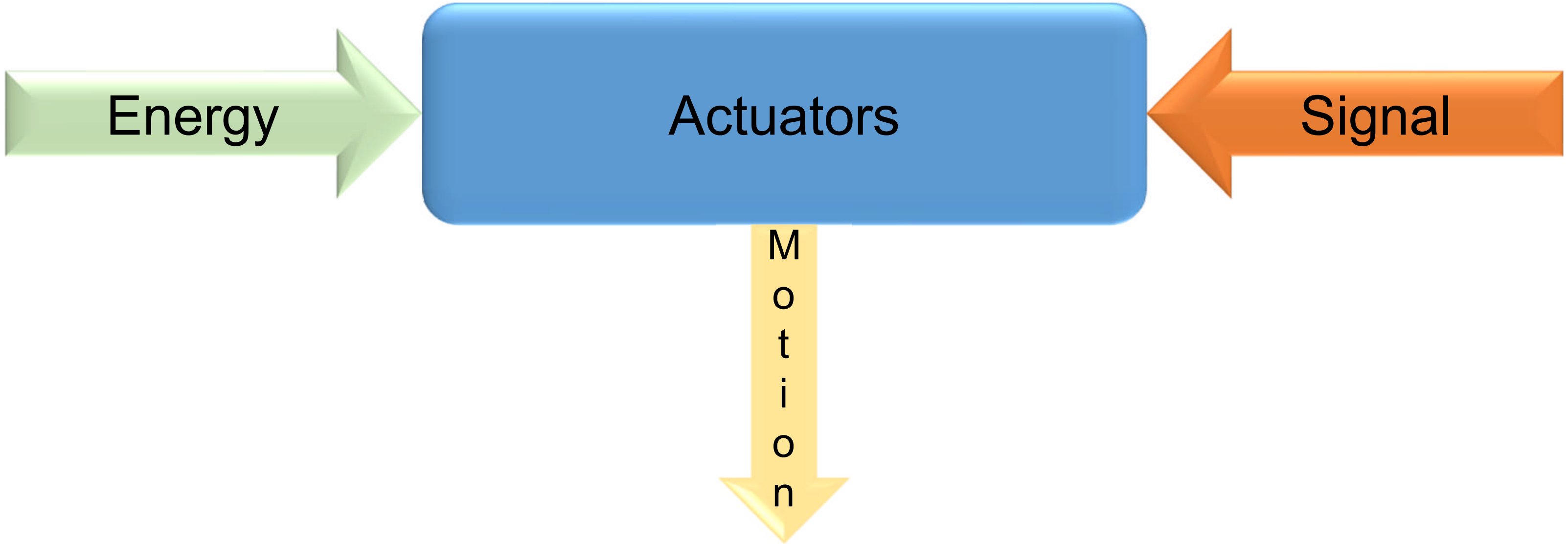
- An ultrasonic sensor is an electronic device that measures the distance of a target object by emitting ultrasonic sound waves, and converts the reflected sound into an electrical signal.
- Ultrasonic waves travel faster than the speed of audible sound

Ultrasonic Sensor – Working Principle

Reference: https://robu.in/ultrasonic-sensor-working-principle/

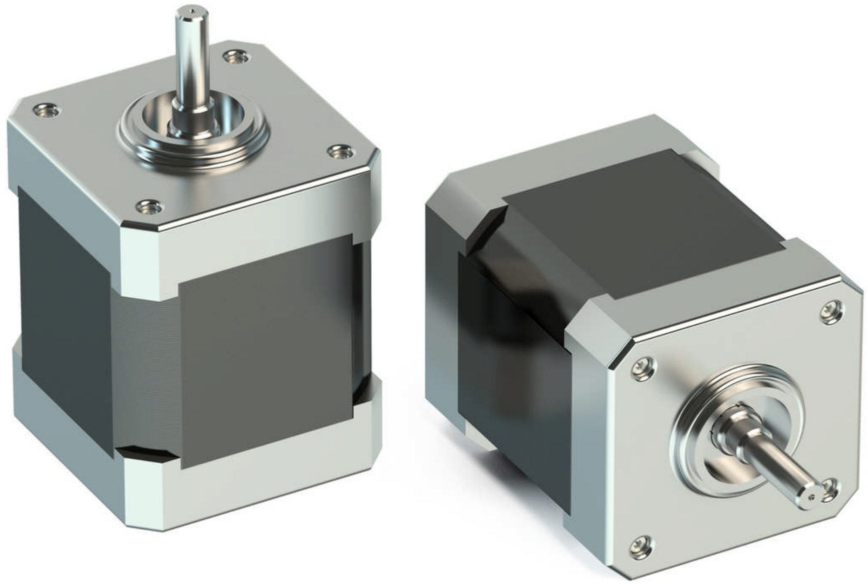
Introduction to Actuators

- An actuator is a device that produces a motion by converting energy and signals going into the system.
- The motion it produces can be either rotary or linear. Linear actuators, as the name implies, produce linear motion.



Stepper Motor

- A **stepper mottor** is a brushless, synchronous electric motor that converts digital pulses into mechanical shaft rotation.
- Every revolution of the stepper motor is divided into a discrete number of steps, in many cases 200 steps, and the motor must be sent a separate pulse for each step.



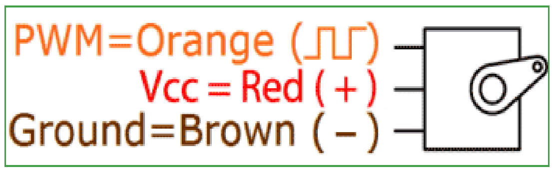
https://www.automate.org/blogs/what-kinds-of-applications-are-best-for-stepper-motors

Stepper Motor - Application

- 3D printing equipment
- Textile machines
- Printing presses
- Gaming machines
- Medical imaging machinery
- Small robotics
- CNC milling machines
- Welding equipment

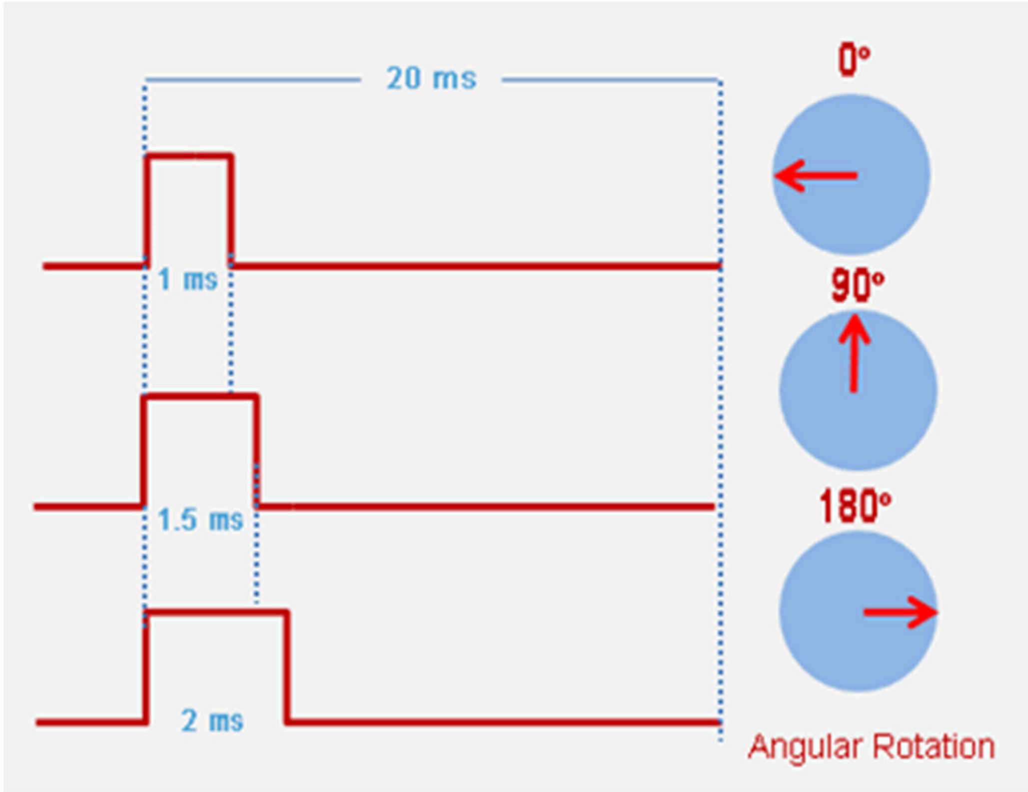
Servo Motor

- A **servo motor** is a type of motor that can rotate with great precision.
- Normally this type of motor consists of a control circuit that provides feedback on the current position of the motor shaft, this feedback allows the servo motors to rotate with great precision.
- A servo motor usually comes with a gear arrangement that allows us to get a very high torque servo motor in small and lightweight packages.



Servo Motor

- Servo motor works on PWM (Pulse width modulation) principle, means its angle of rotation is controlled by the duration of applied pulse to its Control PIN.
- Each 0.5 ms signal move 90 degree rotation



LCD – Liquid Crystal Display

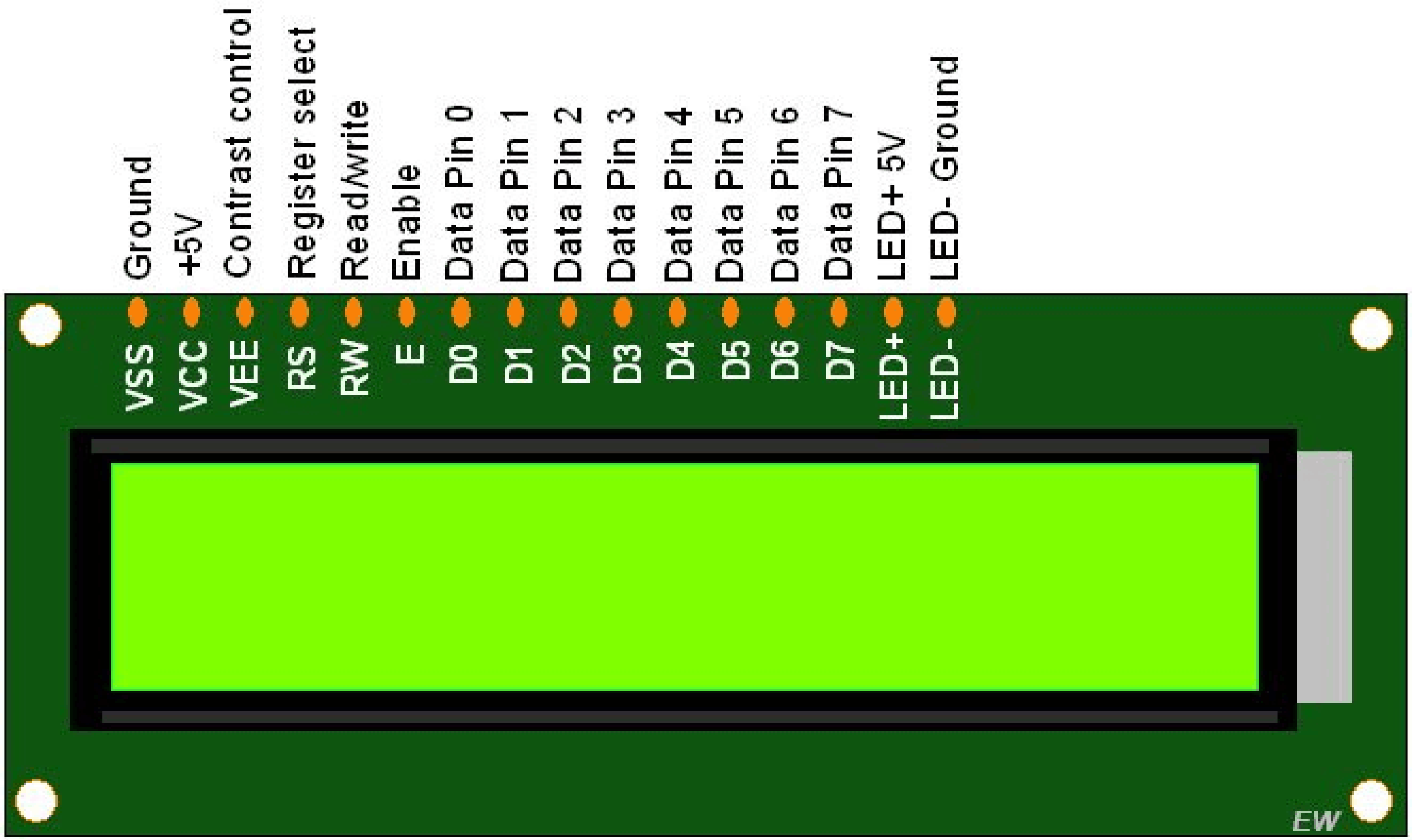
- LCD stands for Liquid Crystal Display that uses a plane panel display technology, used in screens of computer monitors & TVs, smartphones, tablets, mobile devices, etc.

- What is 16x2 LCD?
- An electronic device that is used to display data and the message is known as LCD 16×2.

https://www.watelectronics.com/lcd-16x2/

- As the name suggests, it includes 16 Columns & 2 Rows so it can display 32 characters (16×2=32) in total & every character will be made with 5×8 (40) Pixel Dots. So the total pixels within this LCD can be calculated as 32 x 40 otherwise 1280 pixels.

LCD – Pin Configuration



https://www.electronicwings.com/public/storage/PlatformSection/TopicContent/35/description/1_LCD16x2_Pins.png

Reference

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